

EXPERIMENT NO 3:

Designing a Quarter wave transformer for impedance matching for the following parameters a) Frequency b) Input impedance c) Characteristic impedance

Program:

```
EXP3-TRANSMISSION.sci (c:\users\isra\documents\scilab\EXP3-TRANSMISSION.sci) - Scinotes
EXP1-TRANSMISSION.sci x EXP2-TRANSMISSION.sci x EXP3-TRANSMISSION.sci x
1 clear;clc;
2 Zr=100;s=9;d=0.1;
3 Zs=50;
4 Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;
5 Zof=276*log10(s/r);
6 Zoq=sqrt(Zr*Zof);
7 L=300/(f*(10^-6));
8 printf("length-of-lambda/4-transformer=.%f-meters\n",L);
9 printf("characteristic-impedance-of-quarter-wave-line=.%f-",fix(Ro*10000)/10000);
10
```

OUTPUT:

```
Scilab 2025.1.0 Console
length of lambda/4 transformer = 0.600000 meters
characteristic impedance of quarter wave line = 70.710600
-->
```